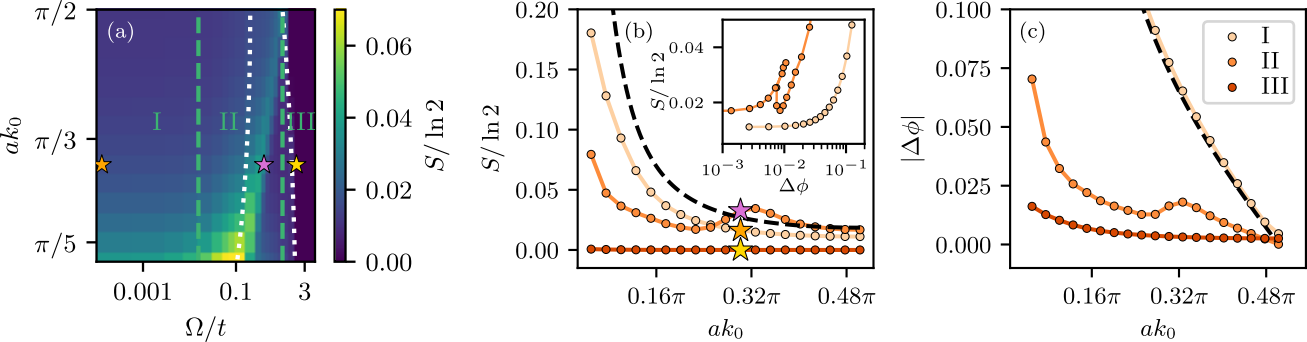

(a)
I
II
III
$\pi/2$
$\pi/3$
$\pi/5$
ak_0
0.001
0.1
3
Ω/t
0.06
0.04
0.02
0.00
$S/\ln 2$
(b)
0.20
0.15
0.10
0.05
0.00
$S/\ln 2$
0.04
0.02
$S/\ln 2$
10^{-3}
10^{-2}
10^{-1}
$\Delta\phi$
0.16π
0.32π
0.48π
ak_0
(c)
0.100
0.075
0.050
0.025
0.000
$|\Delta\phi|$
I
II
III
0.16π
0.32π
0.48π
ak_0